American Computer Science League

2024-2025 ● Contest 3: Short Problems Solutions ● Junior Division

1. Boolean Algebra

$A\overline{B} + \overline{A\overline{B}} + A = A\overline{B} + (\overline{A} + \overline{B}) + A$
$= A\overline{B} + \overline{B} + (\overline{A} + A)$
$= A\overline{B} + \overline{B} + 1$
$= 1$

E. 4

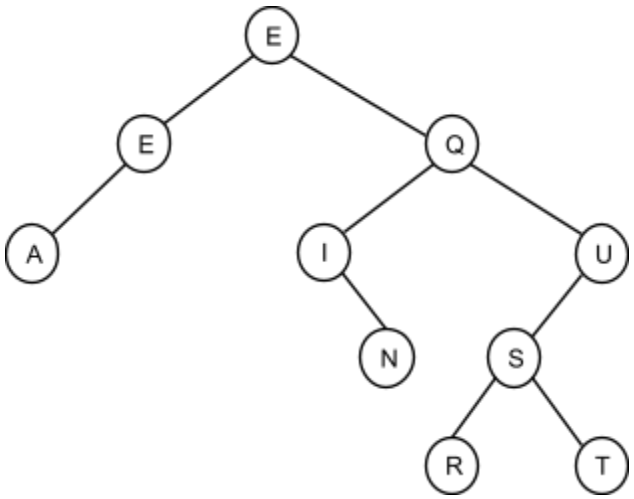
2. Boolean Algebra

$\overline{A\overline{B}} + \overline{A(B + \overline{A})} = \overline{A} + \overline{B} + \overline{A} + \overline{(B + \overline{A})}$
$= \overline{A} + \overline{B} + \overline{A} + \overline{\overline{A}}\,\overline{B}$
$= \overline{A} + \overline{B} + A\overline{B}$
$= \overline{A} + \overline{B}(1 + A)$
$= \overline{A} + \overline{B}$

A. $\overline{A} + \overline{B}$

3. Data Structures

The binary search tree for EQUESTRIAN is:



The nodes with only one child are: E, I, U. There are 3.

C. 3

4. Data Structures

The queue is constructed using FIFO as follows:

F FE FEN EN ENC NC NCI NCIN CIN CING ING NG

Items left from the rear to the front of the queue are: G N

D. G N

5. What Does This Program Do? (Arrays)

arr =

2	3	4	7	6	5	1

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a = arr(3) - arr(5)
b = arr(6) + arr(1)
output arr(a) * arr(b)
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a = 7 −5 = 2,
b = 1 + 3 =4,
arr(2) * arr(4) = 4 * 6 = 24

B. 24

6. What Does This Program Do? (Arrays)

arr -

3	6	7	1	4
1	5	2	3	3
7	4	6	0	3

```
s = 0
for x = 1 to 2
    for y = 0 to 4
        if arr(x,y)<arr(x-1,y) then
            s = s + 1
        end if
    next y
next x
output s
```

This program counts those values in rows 1 and 2 that are less than the values in the previous row. They are 1, 5, 2, 3, 4, 0.

A. 6